Symmetry in Exact diagonalization method

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The central problem of many-body physics is to diagonalize the Hamiltonian $\mathcal{H}$ in the Hilbert space $\mathscr{H}$. To be explicit, in computational basis we have

$$\mathscr{H} = \mathrm{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sigma_i = 1, \dots, d\}. \tag{1}$$

The problem with a naive approach is that $\mathscr{H}$ is exponentially large. Let L be the lattice size and d be the local Hilbert space dimension, the dimension of $\mathscr{H}$ is d^L. If the system has $\mathsf{U}(1)$ symmetry, then $\mathscr{H}$ is further decomposed into subspaces $\mathscr{H}_N$ with fixed particle number N, defined as

$$\mathscr{H}_N = \mathrm{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sum_{i=1}^{L} \sigma_i = N\}, \tag{2}$$

and the dimension of $\mathscr{H}_N$ is $\binom{L}{N}$, which is still exponentially large.

To address such problem, we will use the representation theory[1] to decompose the Hilbert space into irreducible representations of the symmetry group G of the Hamiltonian $\mathcal{H}$.

We will assume G is a finite group, and it acts on the Hilbert space $\mathscr{H}$ via a unitary representation $U : G \to U(\mathscr{H})$. What we mean by $\mathcal{H}$ has symmetry G is that $\mathcal{H}$ commutes with the action of G, i.e. $[U(g), \mathcal{H}] = 0$ for all $g \in G$. By Schur's lemma, we know that $\mathcal{H}$ is block diagonal in the basis of irreducible representations of G. Hence we can diagonalize $\mathcal{H}$ in each irreducible representation separately, which reduces the computational cost significantly.

1 Building the symmetry group

In our case case, we consider two kinds of symmetry: the lattice symmetry that permutes the sites and the internal symmetry that performs a spin flip:

$$g\,|\sigma_1 \dots \sigma_L\rangle = \left|\sigma_{g^{-1}(1)} \dots \sigma_{g^{-1}(L)}\right\rangle, \quad g \in G_{\text{latt}} \tag{3}$$

$$\mathcal{F}\,|\sigma_1 \dots \sigma_L\rangle = |(d-\sigma_1), \dots, (d-\sigma_L)\rangle \tag{4}$$

and the full symmetry group is $G = G_{\text{latt}} \times \mathbb{Z}_2$. The lattice symmetry group G_{latt} is further decomposed into a semi-direct product of the translation group G_{tr} and the point group G_{pt}, i.e. $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$, where $G_{\text{tr}} = \mathbb{Z}_{N_1} \times \cdots \mathbb{Z}_{N_d}$ is a finite Abelian group and G_{pt} is a finite group.

Suppose we already know all the information of G_{pt}, we need to

1. Find the conjugacy classes of G_{latt} and G.
2. Calculate the character table of G_{latt} and G.

1.1 The lattice Group

First we describe the structure of the lattice group $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$. Let $\mathbf{t} \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$. The element in G_{latt} can be written as $(\mathbf{t}, r)$. It acts on the real space as

$$(\mathbf{t}, r)\mathbf{x} = r\mathbf{x} + \mathbf{t}. \tag{5}$$

The group multiplication defined as

$$(\mathbf{t}_1, r_1)(\mathbf{t}_2, r_2) = (\mathbf{t}_1 + r_1\mathbf{t}_2, r_1 r_2) \tag{6}$$

where $r(\mathbf{t})$ is the action of r on $\mathbf{t}$. The inverse of $(\mathbf{t}, r)$ is given by

$$(\mathbf{t}, r)^{-1} = (-r^{-1}\mathbf{t}, r^{-1}). \tag{7}$$

Conjugacy Class Since G_{tr} is a normal subgroup of G_{latt}, the conjugacy class of G_{latt} is divided into two cases:

1. Since

$$(a, r)(\mathbf{t}, 1)(a, r)^{-1} = (a + r\mathbf{t}, r)(-r^{-1}a, r^{-1}) \tag{8}$$

$$= (a + r\mathbf{t} - r^{-1}a, 1) \tag{9}$$

for $(\mathbf{t}, 1)$ and $(\mathbf{t}', 1)$ to be in the same conjugacy class, we need $\mathbf{t}' = a + r\mathbf{t} - r^{-1}a$ for some $a \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$.

2. Translations and point group elements. We have

$$(\mathbf{a}, r)(\mathbf{t}, s)(\mathbf{a}, r)^{-1} = (\mathbf{a} + r\mathbf{t}, rs)(-r^{-1}\mathbf{a}, r^{-1}) \tag{10}$$
$$= (\mathbf{a} + r\mathbf{t} - r^{-1}\mathbf{a}, rsr^{-1}) \tag{11}$$

Hence for $(\mathbf{t}, s)$ and $(\mathbf{t}', s')$ to be in the same conjugacy class, we need $\mathbf{s}$ and $\mathbf{s}'$ to be in the same conjugacy class of G_{pt}, and $\mathbf{t}' = \mathbf{a} + r\mathbf{t} - r^{-1}\mathbf{a}$ for some $\mathbf{a} \in G_{\mathrm{tr}}$ and $r \in G_{\mathrm{pt}}$.

Character The irreducible representations of G_{tr} are one-dimensional, and are labeled by the momentum

$$\mathbf{k} \in \mathrm{BZ} = \left\{ (k_1, \ldots, k_d) \mid k_i = 0, \frac{2\pi}{N_i}, \ldots, \frac{2\pi(N_i - 1)}{N_i} \right\}, \tag{12}$$

with the character defined as

$$\chi_{\mathbf{k}}^{G_{\mathrm{tr}}}(\mathbf{t}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{t}}. \tag{13}$$

The irreducible representations of G_{latt} are labeled by the momentum $\mathbf{k}$ and the irreducible representations of the little group

$$G_{\mathbf{k}} = \{ r \in G_{\mathrm{pt}} \mid r\mathbf{k} = \mathbf{k} \} \subset G_{\mathrm{pt}}. \tag{14}$$

Suppose we know all the irreducible representations of the little group $G_{\mathbf{k}}$, with its character denoted as $\chi_{\alpha}^{G_{\mathbf{k}}}$, then the irreducible representations of G_{latt} are given by

$$\chi_{\mathbf{k},\alpha}^{G_{\mathrm{latt}}}(t, r) = \sum_{g \in G_{\mathrm{pt}}/G_{\mathbf{k}}} \chi_{\mathbf{k}}^{G_{\mathrm{tr}}}(gt)\chi_{\alpha}^{G_{\mathbf{k}}}(grg^{-1}) \tag{15}$$

where g runs over the left cosets of $G_{\mathbf{k}}$ in G_{pt}.

References

[1] William Fulton and Joe Harris. *Representation Theory: A First Course*. Springer, 1991.